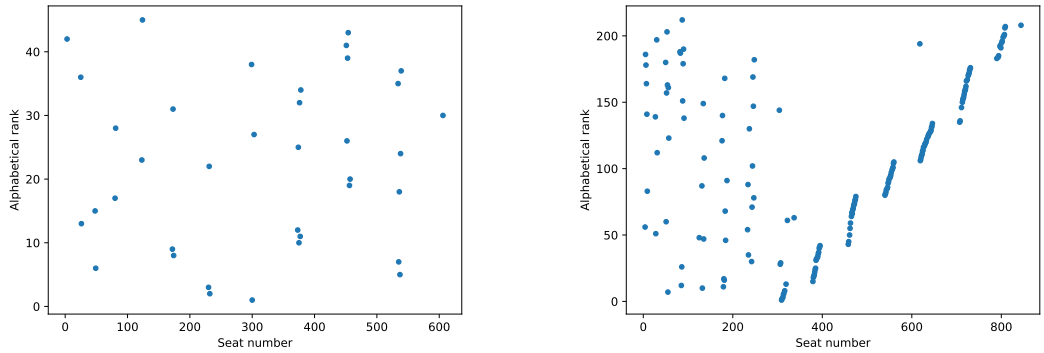
Figure 2.3.2: Seating and alphabetical rank.



Panel A - GUE/NGL (non-alphabetical). *Panel B - S&D (alphabetical seating).*

2.4 Language similarity

2.4.1 Vector embedding

To measure the influence of peer effects on MEPs' speeches, I employ advanced techniques in natural language processing, specifically vector embedding. This approach captures both the similarity in topics chosen by speakers and in the phrases they use to discuss them.

Vector embedding is a cutting-edge method in natural language processing. It shares its technology with word embeddings. Word embedding is an algorithm that generates dense numerical representations of words based on their co-occurrence statistics in large corpora (Mikolov et al. (2013); Pennington et al. (2014)). It maps each word to a lower-dimensional space, where semantically related words are grouped closer together. Words with strong co-occurrence relationships will exhibit similar correlations in this vector space. The model identifies a set of coordinates for each word vector, determining the angles between vectors. By capturing these angles, word embeddings establish meaningful connections between textual elements. For example, words like "economics" and "markets" will be close to each other, whereas "economics" and "constitution" will be farther apart. Additionally, words like "economics" and "economy" will demonstrate even higher similarity, resulting in a higher measured similarity. This word embedding approach provides a continuous measure of semantic distance, addressing the limitations of sparse word counting from a dictionary.

Table 2.3.3: Overview of samples used in the analysis.

	Main analysis Non-leaders, EPGs using alphabetical seating	Placebo sample 1: EPGs not using alphabetical seating	Placebo sample 2: Leaders, EPGs using alphabetical seating	Total available
EP5 (1999-2004)				
Frequencies:				
Nbr of EPGs	4	3	4	7
Nbr of MEPs	463	194	90	747
Nbr of speeches	34,562	6,253	12,166	52,981
MEP characteristics:				
Mean age	54.35	56.06	55.95	55.12
Mean tenure in EP	0.81	1.38	1.14	1.02
Share women	0.34	0.13	0.23	0.23
Share PhD	0.6	0.4	0.9	0.6
Share newcomer	0.44	0.23	0.15	0.34
Share reelected	0.5	0.41	0.73	0.46
EP6 (2004-2009)				
Frequencies:				
Nbr of EPGs	4	3	4	7
Nbr of MEPs	637	674	354	908
Nbr of speeches	39,356	8,568	17,398	65,322
MEP characteristics:				
Mean age	49.86	55.23	52.01	51.12
Mean tenure in EP	0.75	0.86	1.64	0.88
Share women	0.43	0.18	0.39	0.40
Share PhD	0.11	0.08	0.06	0.09
Share newcomer	0.55	0.37	0.16	0.44
Share reelected	0.44	0.31	0.84	0.40
EP7 (2009-2014)				
Frequencies:				
Nbr of EPGs	4	4	4	8
Nbr of MEPs	578	770	26	844
Nbr of speeches	101,232	18,323	30,919	150,474
MEP characteristics:				
Mean age	52.33	54.41	57.69	53.62
Mean tenure in EP	0.55	0.62	1.27	0.61
Share women	0.45	0.21	0.21	0.34
Share PhD	0.17	0.3	0.13	0.14
Share newcomer	0.46	0.49	0.13	0.42
Share reelected	0.44	0.31	0.79	0.37
EP8 (2014-2019)				
Frequencies:				
Nbr of EPGs	4	4	4	8
Nbr of MEPs	504	809	25	850
Nbr of speeches	159,954	23,369	30,866	214,186
MEP characteristics:				
Mean age	48.23	53.57	51.28	51.62
Mean tenure in EP	0.66	0.45	1.33	0.60
Share women	0.46	0.21	0.28	0.35
Share PhD	0.13	0.08	0.057	0.09
Share newcomer	0.57	0.68	0.06	0.60
Share reelected	0.38	0.23	0.82	0.33
EP9 (2019-2022)				
Frequencies:				
Nbr of EPGs	4	4	4	8
Nbr of MEPs	418	725	28	797
Nbr of speeches	26,362	2,325	6,895	35,582
MEP characteristics:				
Mean age	49.24	53.82	50.9	51.22
Mean tenure in EP	0.71	0.81	0.84	0.78
Share women	0.48	0.21	0.28	0.41
Share PhD	0.11	0.09	0.06	0.10
Share newcomer	0.52	0.48	0.55	0.49
Share reelected	0.39	0.34	0.44	0.37

Notes: The table presents counts for various subsets within the dataset. An MEP is classified into one of three groups if they are observed in that group at any given time. As a result, there is some overlap between the groups, and the numbers in the table do not add up to the total count of the entire dataset. The attributes of MEPs are evaluated at the time of their initial recorded speech in the data. The averages and proportions are computed on an individual MEP basis.

One remarkable aspect of vector embedding is the ability to perform algebraic operations on the vectors. The classic example is the equation "$Queen = King + Man - Woman$" which demonstrates the capability to manipulate and reason with the vector representations.

I expand the use of word embeddings by incorporating additional input variables, working at the speech level, in addition to the context words (Le and Mikolov (2014)). This approach, initially referred to as paragraph vectors or party embeddings (Rheault and Cochrane (2019)), allows for a more comprehensive analysis. In this study, the unit of analysis is MEP by session. Just as words have their embeddings, MEPs will possess their own corresponding vectors, effectively representing their language patterns[10] for a given session.

One significant advantage is that these MEP embeddings can be compared against the rest of the corpus vocabulary in a shared vector space, facilitating straightforward comparisons.

I use MEP-session pairs as indicator variables because considering combinations of MEPs and time periods allows to represent language dynamics depending on the discussed laws (and thus topics) on specific dates. Each MEP within a given session has a unique embedding. This approach captures the possibility that the language and issues debated by each MEP may evolve from one session to the next, reflecting the dynamic nature of peer influence. MEP language representations can "move" over time in the vector space.

Among the various vector embedding algorithms and model training options available, I selected the algorithm developed by Mikolov et al. (2013) and Pennington et al. (2014) called *Doc2Vec*. The implementation uses Python's gensim library with three-hundred-dimensional vectors, an eight-object context window, and training for ten rounds. These choices align with standard hyperparameters commonly used in applied natural language processing research. Previous research has shown that in social-science contexts, downstream measurements are not highly sensitive to these choices (Rodriguez and Spirling (2022) and Ash and Weelden (2017)).

A particularity of this corpus is that it gathers the 24 official languages of the European Union since MEPs are allowed to speak in their native language. Therefore, the corpus

[10] *Language pattern* can be understood as a mathematical representation of how a particular MEP communicates or expresses themselves in a given session.

was translated into English when needed using *Google Translate*. As demonstrated by Gao et al. (2023), Li et al. (2019) and Artetxe and Schwenk (2019), automated translation typically maintains consistency and accuracy, especially when using *Doc2Vec*, minimizing potential biases or distortions.

In summary, vector embeddings enable me to capture the intricacies and contextual use of language in MEPs' speeches, facilitating a more comprehensive analysis of peer effects and their impact on language patterns.

2.4.2 Assessing distances

With the vector representations of MEPs, denoted as $\boldsymbol{\zeta}$, I can measure the language similarity between two MEPs i and j by calculating the Euclidean distance $d_{i,j}$ between their corresponding vectors $\boldsymbol{\zeta}_i$ and $\boldsymbol{\zeta}_j$. These vectors, normalized and placed in an M-dimensional space, represent their embedded political speech. ζ_{im} and ζ_{jm} are the m-th components of the vectors for MEPs i and j, respectively. A smaller Euclidean Mean distances greater language similarity within a given session. The distance is computed using the *Doc2Vec* training process, and is expressed as:

$$d_{i,j} = \sqrt{\sum_{m=1}^{M} (\zeta_{im} - \zeta_{jm})^2}$$

The Euclidean distance measure is widely used in linguistics and data mining due to its computational efficiency (Sebastiani (2002)).[11] Economists have also started adopting this measure (Hoberg and Phillips (2016)). In this context, the measures are validated by examining whether pairs of MEPs sharing common characteristics (such as belonging to the same political group or coming from the same member state) exhibit lower Euclidean distance compared to pairs without these shared characteristics (Table 2.5.1). The Euclidean distance is found to be significantly lower for same-group and same-country pairs compared to cross-characteristic pairs, at the 5% or 1% significance level. The difference in Euclidean distance between same-group and same-country pairs and cross-group and cross-country pairs ranges from 0.21 to 0.37 standard deviations of

[11] Another widely used measure is cosine similarity. It should be noted that Euclidean distance and *cosine similarity* become proportional when the vectors are normalized. A demonstration can be found here.

the outcome. This suggests that the language similarity measures capture the partisan nature of speech observed (Gentzkow et al. (2019b)).

2.5 Empirical strategy and results

2.5.1 Empirical strategy

To investigate the potential influence of physical proximity on speaking behaviour, I examine the Euclidean distance among pairs of MEPs who are seated next to each other. First, the main analysis focuses on each session within each EPG and considers all possible pairs of MEPs within the same EPG who are present and participate from September 2004 until June 2019. In this section, I adopt the empirical strategy used by Harmon et al. (2019). Thus, data is at the MEP pair-session level, with one observation for each possible MEP pair in any given session. With N MEPs in a given session, this means a total of $\frac{N(N-1)}{2}$ observations for that session. Using the indexing notation, ij represents the MEP pairs, and t represents the specific session. The variable $Distance_{ijt}$ represents the normalized Euclidean distance of speaking between individual i and individual j during time t. Additionally, the variable $SeatNeighbours_{ijt}$ is created to indicate if MEP pair ij were seated side by side during the respective session t.

To estimate the causal impact of sitting together on speaking concordance, a naive approach would compare the speaking behaviour of MEP pairs who sit adjacent to each other (treated observations) with those who do not (untreated observations). This is achieved by regressing the dependent variable, $Distance_{ijt}$ on the explanatory variable, $SeatNeighbours_{ijt}$:

$$Distance_{ijt} = \beta_0 + \beta_1 SeatNeighbours_{ijt} + \nu_{ijt} \tag{2.5.1}$$

Considering the estimate of β_1 as the causal impact of sitting side by side on speaking behaviour raises two concerns. Firstly, MEPs who already exhibit similar speaking behaviour might be more inclined to $decide$ to seat next to each other. Secondly, even among MEPs who adhere to the alphabetical seating rule, the rule itself may prompt MEPs with a higher likelihood of having similar speaking behaviour to choose to sit